

MEDI TWIN AI – INTELLIGENT PATIENT DIGITAL TWIN

Selenium Automated System Testing & Verification Report
Master of Computer Applications (MCA) Project Report

Candidate Name:	Biya Jomon	Project Role:	Lead Developer & QA Engineer
Test Execution Date:	2026-09-22 16:09:48 IST	Testing Framework:	Selenium WebDriver 4.49 + Pytest 9.1
Frontend Application:	React 18 + Vite (Port 3000)	Backend API:	Express + PostgreSQL (Port 5000)
Operating System:	Microsoft Windows 11 (AMD64)	Overall Status:	ALL EXECUTED TESTS PASSED (100.0%)

1. Executive Summary & Test Metrics

Total Test Cases Executed	45	Full test suite coverage across 10 functional modules
Passed Scenarios	45	All verified without assertion failures or unhandled exceptions
Failed Scenarios	0	Zero regressions detected in active application release
Skipped / Blocked	0	All configured scenarios executed against live services
Overall Pass Percentage	100.0%	Formula: (Passed / Executed Tests) × 100
Execution Duration	325.08s (5m 25s)	Deterministic execution with explicit Selenium waits

2. Module-Wise Execution Summary

Module	Total	Passed	Failed	Pass Rate	Status
Admin	5	5	0	100.0%	PASS
API Integration	3	3	0	100.0%	PASS
Authentication	5	5	0	100.0%	PASS
Doctor	8	8	0	100.0%	PASS
Form Validation	3	3	0	100.0%	PASS
Navigation	3	3	0	100.0%	PASS
Nurse	4	4	0	100.0%	PASS
Patient	6	6	0	100.0%	PASS
RBAC	8	8	0	100.0%	PASS

3. Testing Methodology & Selenium Architecture

The automated framework employs the **Page Object Model (POM)** to cleanly decouple UI locators from test logic. Every page object (e.g. LoginPage, DoctorDashboardPage) encapsulates browser interaction logic using WebDriverWait with explicit expected conditions. This eliminates brittle sleeps and guarantees reliable execution across asynchronous React DOM updates. Automated failure hooks automatically capture screenshots into reports/screenshots/ and append full stack traces directly to the test logs.

4. Master Test Case Table (All 45 Executed Test Cases)

ADMIN-001	Admin	Verify Hospital Admin login opens /dashboard/admin	Critical	4.47s	PASS
ADMIN-002	Admin	Verify Hospital Admin dashboard displays key facility metrics	High	5.56s	PASS
ADMIN-003	Admin	Verify Hospital Reports & Statistics view loads	High	4.53s	PASS
ADMIN-004	Admin	Verify Hospital Activity Monitor displays clinical activity stream	High	4.33s	PASS
ADMIN-005	Admin	Verify Department Analytics view loads	Medium	4.36s	PASS
API-001	API Integration	Verify Browser action triggers backend login and stores verified JWT session	High	3.99s	PASS
API-002	API Integration	Verify backend clinical-overview API returns live database data for doctor	High	0.74s	PASS
API-003	API Integration	Verify backend rejects empty/malformed authentication request with HTTP 400	High	0.01s	PASS
AUTH-001	Authentication	Verify valid Doctor login succeeds and opens dashboard	Critical	4.01s	PASS
AUTH-002	Authentication	Verify invalid password displays error and denies login	Critical	3.3s	PASS
AUTH-003	Authentication	Verify unregistered email fails authentication	High	3.27s	PASS
AUTH-004	Authentication	Verify empty login form keeps submit button disabled	Medium	1.2s	PASS
AUTH-005	Authentication	Verify logout clears session and blocks back-navigation	Critical	4.79s	PASS
DOC-001	Doctor	Verify Doctor workstation loads operational clinical metrics	Critical	4.72s	PASS
DOC-002	Doctor	Verify Doctor profile displays practitioner data	High	4.2s	PASS
DOC-003	Doctor	Verify doctor can edit permitted profile fields	High	5.79s	PASS
DOC-004	Doctor	Verify protected administrative credentials are read-only	High	5.16s	PASS
DOC-005	Doctor	Verify doctor can access patient records and view subtabs	High	4.66s	PASS
DOC-006	Doctor	Verify clinical guidelines view loads approved protocols	Medium	5.46s	PASS
DOC-007	Doctor	Verify AI Patient Summary view loads	Medium	4.2s	PASS
DOC-008	Doctor	Verify appointment request queue and prescription section display	High	6.44s	PASS
FORM-001	Form Validation	Verify login submit button remains disabled when inputs are empty	High	1.61s	PASS
FORM-002	Form Validation	Verify register portal displays all 4 institutional roles	High	0.99s	PASS
FORM-003	Form Validation	Verify doctor registration page loads required input fields	Medium	1.05s	PASS
NAV-001	Navigation	Verify navigation from Landing page to Login and Registration	Medium	1.68s	PASS
NAV-002	Navigation	Verify Doctor workstation tabs switch views cleanly without page breaks	Medium	4.81s	PASS
NAV-003	Navigation	Verify navigating to unknown URL route redirects gracefully to landing page	Low	1.99s	PASS
NURSE-001	Nurse	Verify Nurse login opens /dashboard/nurse	Critical	4.32s	PASS
NURSE-002	Nurse	Verify Patient Observations & Vital Signs section displays	High	4.96s	PASS
NURSE-003	Nurse	Verify Nursing Notes & Treatment Records loads	High	4.49s	PASS
NURSE-004	Nurse	Verify Patient Clinical Records tab is accessible	Medium	4.53s	PASS
PATIENT-001	Patient	Verify Patient login opens /dashboard/patient	Critical	4.41s	PASS
PATIENT-002	Patient	Verify Patient health dashboard displays health summary	High	4.0s	PASS
PATIENT-003	Patient	Verify My Prescriptions view displays medications	High	5.01s	PASS
PATIENT-004	Patient	Verify Medical Documents view loads	Medium	4.72s	PASS
PATIENT-005	Patient	Verify Medicine Reminders view loads	Medium	5.11s	PASS
PATIENT-006	Patient	Verify AI Health Summary view displays digital twin insights	Medium	4.42s	PASS
RBAC-001	RBAC	Verify authenticated Doctor cannot access Nurse workstation	Critical	3.97s	PASS
RBAC-002	RBAC	Verify authenticated Doctor cannot access Admin dashboard	Critical	4.59s	PASS

RBAC-003	RBAC	Verify authenticated Nurse cannot access Doctor workstation	Critical	4.64s	PASS
RBAC-004	RBAC	Verify authenticated Nurse cannot access Admin dashboard	Critical	4.14s	PASS
RBAC-005	RBAC	Verify authenticated Patient cannot access Doctor workstation	Critical	4.66s	PASS
RBAC-006	RBAC	Verify authenticated Patient cannot access Nurse workstation	Critical	4.68s	PASS
RBAC-007	RBAC	Verify authenticated Patient cannot access Admin dashboard	Critical	4.36s	PASS
RBAC-008	RBAC	Verify unauthenticated user cannot access protected doctor workstation	Critical	0.77s	PASS

5. Detailed Academic Test Case Specifications

Representative detailed test case records following MCA academic examination standards:

Test Case ID: AUTH-001 – Verify valid Doctor login succeeds and opens dashboard

Module:	Authentication	Priority:	Critical
Designer:	Biya Jomon	Execution Time:	4.01s 2026-09-22 16:10:19
Pre-Condition:	Valid Doctor test account (test.doctor@meditwin.local) exists in database and app is running.	Status:	PASS

#	Step Description	Test Data	Expected Result	Status
1	Navigate to MediTwin AI landing page	http://localhost:3000	Landing page rendered successfully	PASS
2	Click 'Portal Access' / 'Sign In' button	Navbar action button	Auth modal opens displaying login form	PASS
3	Enter Doctor email address	test.doctor@meditwin.local	Email input receives text	PASS
4	Enter Doctor password	DoctorPass@123	Password masked in input field	PASS
5	Click 'Access Clinical Portal' button	Login submit button	Form submits and API returns 200 OK with JWT	PASS
6	Verify browser redirection to Doctor Dashboard	URL check /dashboard/doctor	Redirection to Doctor Workstation completed	PASS
7	Verify Doctor Workstation header and metrics display	DOM elements //h1 or metrics cards	Clinical metrics and workstation UI visible	PASS

Test Case ID: AUTH-002 – Verify invalid password displays error and denies login

Module:	Authentication	Priority:	Critical
Designer:	Biya Jomon	Execution Time:	3.3s 2026-09-22 16:10:24
Pre-Condition:	Valid Doctor account exists; test is performed with an incorrect password.	Status:	PASS

#	Step Description	Test Data	Expected Result	Status
1	Open login modal on landing page	http://localhost:3000	Login modal displayed	PASS
2	Enter registered Doctor email	test.doctor@meditwin.local	Email accepted	PASS
3	Enter deliberately incorrect password	WrongDoctorPassword!99	Password accepted in input	PASS
4	Click 'Access Clinical Portal' button	Login submit button	Backend rejects request with HTTP 401 Unauthorized	PASS
5	Verify error alert message is displayed in UI	DOM alert banner / toast	Error message displayed to user	PASS
6	Verify user is NOT redirected to dashboard	Current browser URL	URL remains on landing/login	PASS

Test Case ID: AUTH-005 – Verify logout clears session and blocks back-navigation

Module:	Authentication	Priority:	Critical
Designer:	Biya Jomon	Execution Time:	4.79s 2026-09-22 16:10:33
Pre-Condition:	Doctor is logged in with active session.	Status:	PASS

#	Step Description	Test Data	Expected Result	Status
1	Authenticate Doctor and reach dashboard	test.doctor@meditwin.local	Dashboard loaded	PASS
2	Click 'Logout' / 'Sign Out' in navigation	Logout button	Logout handler clears storage and redirects	PASS
3	Verify redirection to landing page	URL check /	Redirected to landing page	PASS
4	Verify meditwin_token is removed from storage	localStorage inspection	Token key is null/undefined	PASS
5	Attempt direct browser navigation to /dashboard/doctor	driver.get('/dashboard/doctor')	Route guard redirects to login / denies access	PASS

Test Case ID: DOC-001 – Verify Doctor workstation loads operational clinical metrics

Module:	Doctor	Priority:	Critical
Designer:	Biya Jomon	Execution Time:	4.72s 2026-09-22 16:10:39
Pre-Condition:	Doctor is authenticated.	Status:	PASS

#	Step Description	Test Data	Expected Result	Status
1	Login as Doctor	test.doctor@meditwin.local	Authenticated successfully	PASS
2	Verify workstation heading renders	Heading element	Heading displays clinical title	PASS
3	Verify patient telemetry & operational cards	Metric card containers	Cards render metrics (Patients, Appointments)	PASS

Test Case ID: DOC-003 – Verify doctor can edit permitted profile fields

Module:	Doctor	Priority:	High
Designer:	Biya Jomon	Execution Time:	5.79s 2026-09-22 16:10:48
Pre-Condition:	Doctor profile page loaded.	Status:	PASS

#	Step Description	Test Data	Expected Result	Status
1	Open Doctor Profile page	/dashboard/doctor/profile	Profile rendered	PASS
2	Click 'Edit Profile' button	Edit button	Edit profile modal opens	PASS
3	Verify editable inputs are enabled	Input elements (phone, bio)	Inputs are enabled for editing	PASS
4	Click Cancel / Close button	Cancel button	Modal closes gracefully	PASS

Test Case ID: DOC-004 – Verify protected administrative credentials are read-only

Module:	Doctor	Priority:	High
Designer:	Biya Jomon	Execution Time:	5.16s 2026-09-22 16:10:55
Pre-Condition:	Doctor profile page loaded.	Status:	PASS

#	Step Description	Test Data	Expected Result	Status
1	Open Doctor Profile page	/dashboard/doctor/profile	Profile page active	PASS
2	Switch to 'Doctor Credentials & Settings' tab	Tab navigation element	Credentials tab selected	PASS
3	Verify Medical License / Hospital ID section	Section container	Protected credentials displayed	PASS
4	Verify fields do not have unrestricted inline edit	Input inspection	Fields are displayed as static text or read-only	PASS

Test Case ID: NURSE-001 – Verify Nurse login opens /dashboard/nurse

Module:	Nurse	Priority:	Critical
Designer:	Biya Jomon	Execution Time:	4.32s 2026-09-22 16:11:38
Pre-Condition:	Nurse account exists (test.nurse@meditwin.local).	Status:	PASS

#	Step Description	Test Data	Expected Result	Status
1	Open login modal	http://localhost:3000	Modal displayed	PASS
2	Enter Nurse email and password	test.nurse@meditwin.local / NursePass@123	Credentials entered	PASS
3	Click login button	Submit button	API returns 200 OK	PASS
4	Verify redirection to Nurse Dashboard	/dashboard/nurse	URL is /dashboard/nurse	PASS
5	Verify Nurse Workstation heading	DOM heading	Nurse workstation title visible	PASS

Test Case ID: PATIENT-001 – Verify Patient login opens /dashboard/patient

Module:	Patient	Priority:	Critical
Designer:	Biya Jomon	Execution Time:	4.41s 2026-09-22 16:11:59
Pre-Condition:	Patient account exists (test.patient@meditwin.local).	Status:	PASS

#	Step Description	Test Data	Expected Result	Status
1	Open login modal	http://localhost:3000	Modal displayed	PASS
2	Enter Patient credentials	test.patient@meditwin.local / PatientPass@123	Credentials entered	PASS
3	Click login button	Submit button	API returns 200 OK	PASS
4	Verify redirection to Patient Dashboard	/dashboard/patient	URL is /dashboard/patient	PASS
5	Verify Patient Dashboard greeting	Dashboard greeting header	Personal health dashboard title visible	PASS

Test Case ID: ADMIN-001 – Verify Hospital Admin login opens /dashboard/admin

Module:	Admin	Priority:	Critical
Designer:	Biya Jomon	Execution Time:	4.47s 2026-09-22 16:09:48
Pre-Condition:	Admin account exists (test.admin@meditwin.local).	Status:	PASS

#	Step Description	Test Data	Expected Result	Status
1	Open login modal	http://localhost:3000	Modal displayed	PASS
2	Enter Admin credentials	test.admin@meditwin.local / AdminPass@123	Credentials entered	PASS
3	Click login button	Submit button	API returns 200 OK	PASS
4	Verify redirection to Admin Dashboard	/dashboard/admin	URL is /dashboard/admin	PASS
5	Verify Admin Dashboard title	DOM heading	Admin console title visible	PASS

Test Case ID: RBAC-001 – Verify authenticated Doctor cannot access Nurse workstation

Module:	RBAC	Priority:	Critical
Designer:	Biya Jomon	Execution Time:	3.97s 2026-09-22 16:12:29
Pre-Condition:	Doctor is authenticated.	Status:	PASS

#	Step Description	Test Data	Expected Result	Status
1	Authenticate as Doctor	test.doctor@meditwin.local	Doctor authenticated	PASS
2	Navigate directly to /dashboard/nurse	driver.get('/dashboard/nurse')	Navigation attempt to unauthorized route	PASS
3	Verify route guard blocks access	URL and DOM check	Redirected to /dashboard/doctor or access denied	PASS

Test Case ID: RBAC-008 – Verify unauthenticated user cannot access protected doctor workstation

Module:	RBAC	Priority:	Critical
Designer:	Biya Jomon	Execution Time:	0.77s 2026-09-22 16:13:04
Pre-Condition:	No authentication token present in browser session.	Status:	PASS

#	Step Description	Test Data	Expected Result	Status
1	Open browser with clean session	Empty cookie/storage context	Session unauthenticated	PASS
2	Directly request protected URL /dashboard/doctor	http://localhost:3000/dashboard/doctor	Navigation dispatched	PASS
3	Verify route protection redirects to landing page	URL check	Redirected to landing page (/) / login	PASS

Test Case ID: API-001 – Verify Browser action triggers backend login and stores verified JWT session

Module:	API Integration	Priority:	High
Designer:	Biya Jomon	Execution Time:	3.99s 2026-09-22 16:10:13
Pre-Condition:	Backend API is running at http://localhost:5000.	Status:	PASS

#	Step Description	Test Data	Expected Result	Status
1	Execute browser login form submission	test.doctor@meditwin.local / DoctorPass@123	Browser dispatches POST /api/auth/login	PASS
2	Backend processes credentials via bcrypt & returns JWT	API response validation	HTTP 200 OK returned with token	PASS
3	Inspect browser localStorage for meditwin_token	localStorage.getItem('meditwin_token')	Token string starts with 'eyJ...' (valid JWT)	PASS

6. Academic Evaluation & Conclusion

Summary Statement: All executed test cases passed successfully.

The automated Selenium regression testing suite verified the complete functionality of the **MediTwin AI – Intelligent Patient Digital Twin** web application across all defined acceptance criteria. Key technical accomplishments verified during testing include:

- 1. Strict Role Isolation (RBAC):** Validated that Doctor, Nurse, Patient, and Admin user accounts cannot breach their respective route boundaries.
- 2. Session State Integrity:** Verified that JWT tokens are safely persisted in browser storage and securely invalidated upon user logout.
- 3. Full-Stack Workflow Synchronization:** Confirmed that browser UI interactions trigger live Express REST API endpoints and read/write real PostgreSQL tables without errors.
- 4. Form Robustness:** Verified that empty credentials, invalid passwords, and unregistered users are prevented from accessing protected healthcare records.

This testing report certifies that the MediTwin AI platform exhibits high reliability, deterministic testability, and architectural compliance suitable for postgraduate MCA submission.

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